

# GEOGRAPHY

## PHYSICAL GEOGRAPHY



## MODULE -2

- BIG BANG THEORY
- UNIVERSE, STARS & PLANETS
- TERRESTRIAL PLANETS
- JOVIAN PLANETS

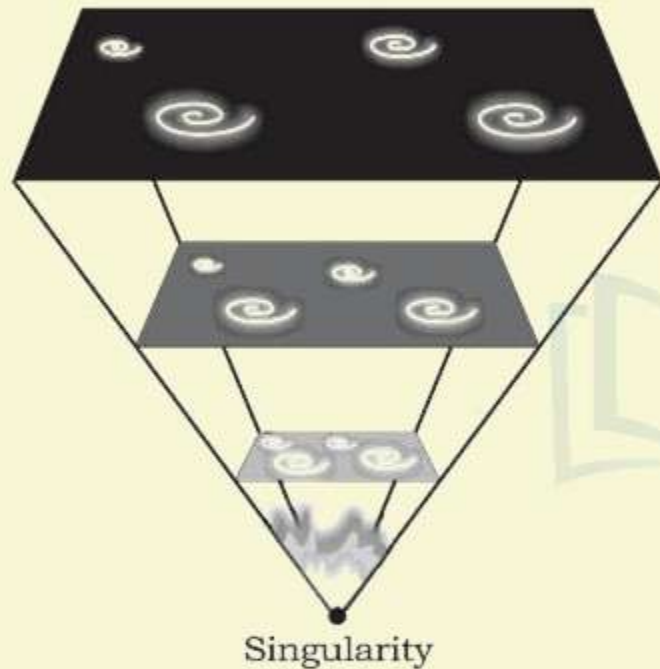




# UNIVERSE & EVOLUTION OF EARTH



## BIG BANG THEORY



- All matter forming the universe existed in one place in the form of a tiny ball (singular atom) with unimaginably small volume, infinite temperature and infinite density.
- At the Big Bang, the 'tiny ball' exploded violently.
- This led to huge expansion.
- It took place around 13.7 billion years before the present.
- The expansion continues even to the present day.
- As it grew, some energy was converted into matter.

***Within 3,00,000 years from the Big Bang, temperature dropped to 4,500K and gave rise to atomic matter and the expansion of universe means increase in space between the galaxies.***





## FORMATION OF GALAXIES



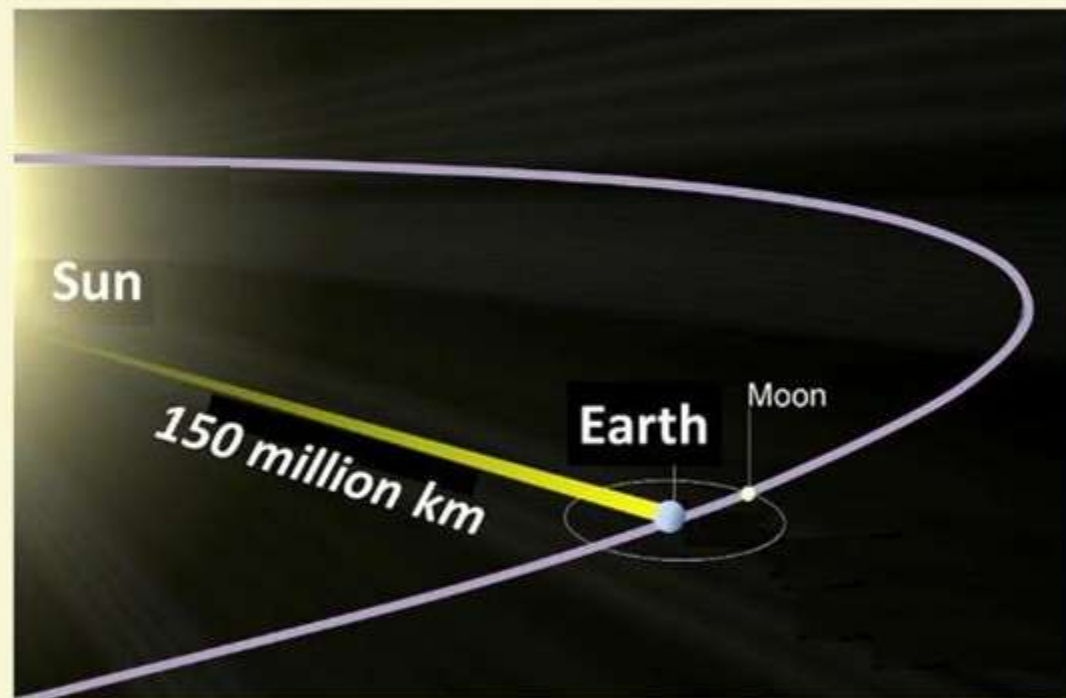
- 1** Distribution of matter and energy was not even in the early universe.
- 2** There were density differences initially.
- 3** It gave rise to differences in gravitational forces and it caused the matter to get drawn together.
- 4** These formed the bases for development of galaxies.
- 5** Galaxies spread over vast distances and are measured in thousands of light-years.
- 6** A galaxy contains a large number of stars.
- 7** The diameters of individual galaxies range from 80,000 to 1,50,000 light years.
- 8** A galaxy starts to form by accumulation of hydrogen gas in the form of a very large cloud called nebula.







## WHAT DO YOU UNDERSTAND BY LIGHT YEAR?



- 1 It is a measure of distance.
- 2 Light travels at a speed of  $3 \times 10^8$  m/s or 3,00,000 km/s.
- 3 The distance light travels in one year is taken as one light year.
- 4 This equals to  $9.461 \times 10^{12}$  km.
- 5 The mean distance between the Sun and the Earth is 149,598,000 km.
- 6 In terms of light years, it is 8.311 minutes of a year.





# UNIVERSE



UNIVERSE



BILLIONS OF GALAXIES



BILLIONS OF STARS / PLANETS  
/ CLOUDS OF GAS AND DUST.  
ONE EXAMPLE IS OUR SOLAR  
SYSTEM

**DO YOU KNOW?**

Andromeda Galaxy is the  
nearest large galaxy to us.







# MILKY WAY GALAXY



## WHY OUR GALAXY IS CALLED MILKY WAY GALAXY?

It appears as a **dim glowing band** across the night sky, where it is not possible to distinguish individual stars.

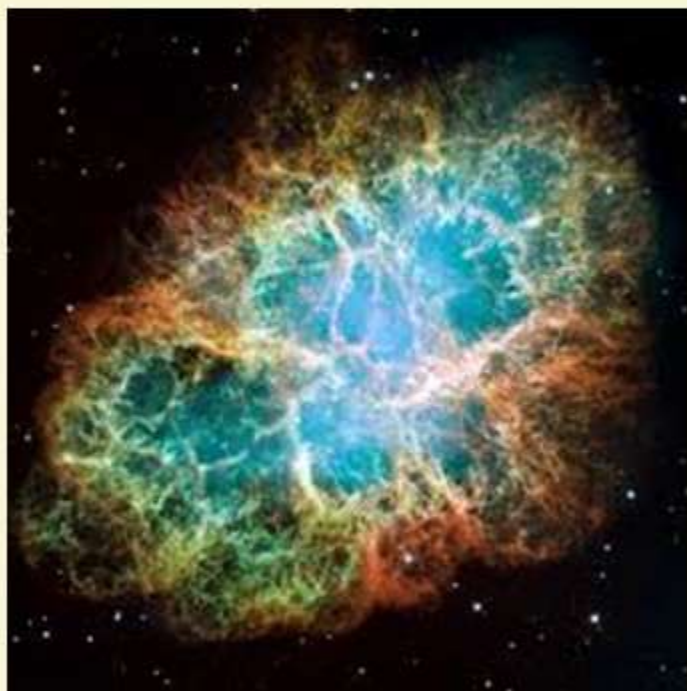
## GALAXY MEANS ...

Huge system of billions of stars, the clouds of dust and gases.



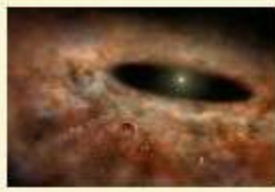


# HOW THE STARS ARE FORMED?



- 1 The growing nebula develops localised **clumps of gas**.
- 2 These clumps continues to grow into even denser gaseous bodies giving rise to formation of stars.
- 3 The formation of stars is believed to have taken place 5 to 6 billion years ago.





# FORMATION OF PLANETS



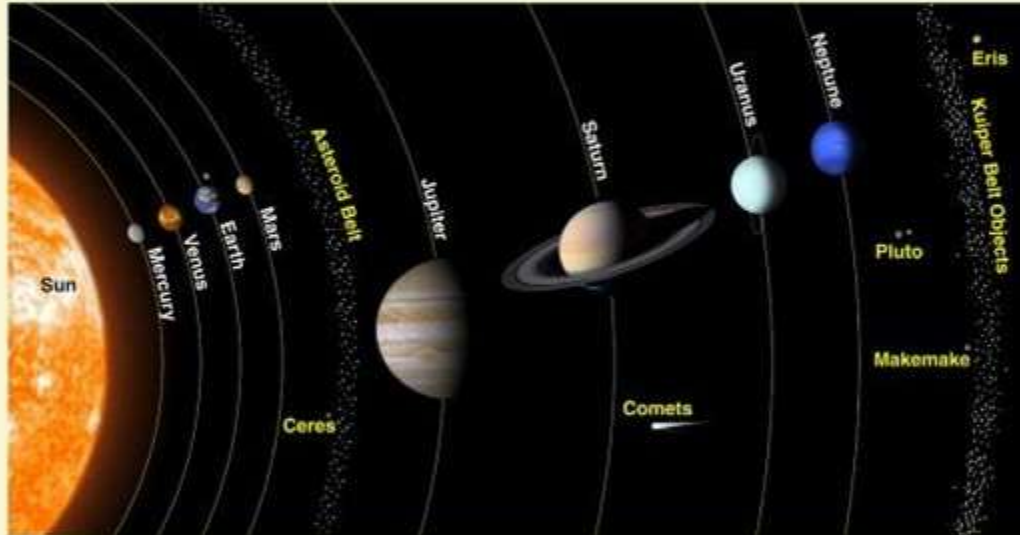
- 1** The stars are localised **lumps of gas** within a nebula.
- 2** The gravitational force within the lumps leads to the formation of a core to the gas cloud and a huge rotating disc of gas and dust develops around the gas core.
- 3** In the next stage, the gas cloud starts getting condensed and the matter around the core develops into small rounded objects.
- 4** These rounded objects by the process of cohesion develop into **planetesimals**.
- 5** Larger bodies start forming by collision and gravitational attraction causes the material to stick together.
- 6** Planetesimals are a large number of smaller bodies.
- 7** These planetesimals finally form fewer large bodies in the form of planets.







# WHAT ABOUT OUR SOLAR SYSTEM?



**1** The nebula from which our solar system is supposed to have been formed started its collapse and core formation some 5 to 5.6 billion years ago and the planets were formed about 4.6 billion years ago.

**2** Our solar system consists of the Sun (the star), 8 planets, moons and millions of smaller bodies like asteroids and comets and huge quantity of dust-grains and gases.





# SOLAR SYSTEM

- Formed 4.6 billion years ago.
- Due to the gravitational collapse of giant molecular cloud.
- Most of the mass is in the Sun.
- It consists of 8 planets, dwarf planets and other celestial objects.
- Distance between Earth and Sun is approx. 150 million km.
- It is part of **MILKY WAY GALAXY**.



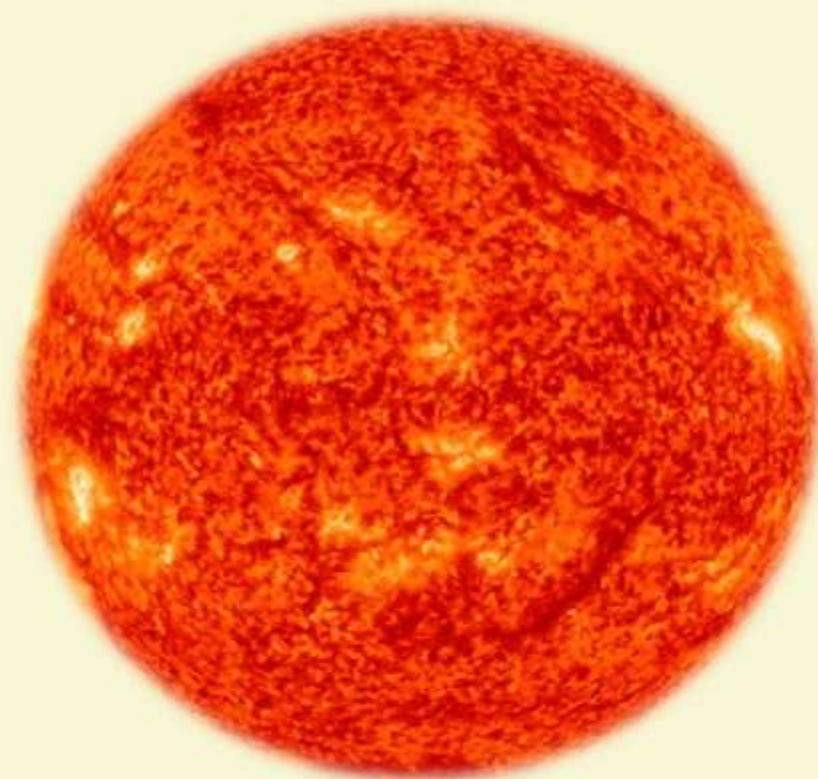






# SUN

- 1 It is made up of extremely hot gases - Hydrogen 71%, Helium 26.5%.
- 2 It weighs  $2 \times 10^{27}$  tonnes and accounts for 99.85% of mass of the solar system.
- 3 The light from the sun reaches the Earth in around 8 minutes.
- 4 It continuously gives off energy in the form of electromagnetic radiation.
- 5 Core of the Sun has got a temperature of 15 million kelvin.
- 6 The immense gravitational pull of the Sun keeps the planets rotating around it in definite orbits.

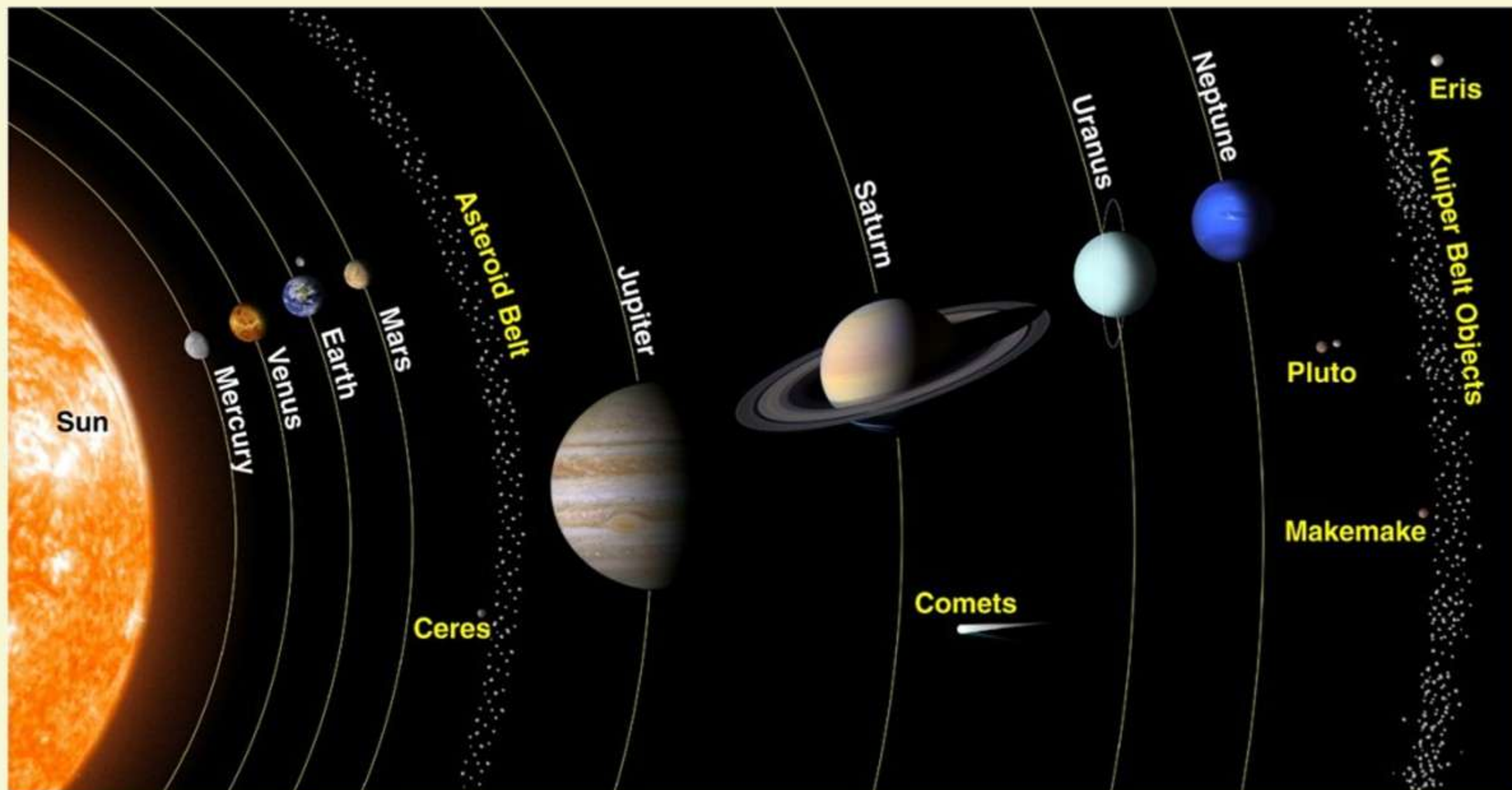


## DO YOU KNOW?

**Alpha Centauri is the nearest star to the Earth except Sun.**

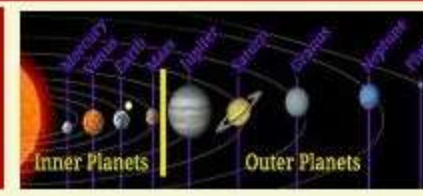








# INNER PLANETS AND OUTER PLANETS



## TERRESTRIAL PLANETS

- They lie between the Sun and the belt of asteroids.
- These are Mercury, Venus, Earth and Mars.
- They are made up of rock and metals.
- They are called Terrestrial because they are Earth-like.
- They have relatively high densities.

## JOVIAN OR GAS GIANTS

- They are much larger than the terrestrial planets.
- They have thick atmosphere, mostly consisting of helium and hydrogen.
- All the planets were formed in the same period sometime about 4.6 billion years ago.







# THE DIFFERENCE BETWEEN TERRESTRIAL & JOVIAN PLANETS



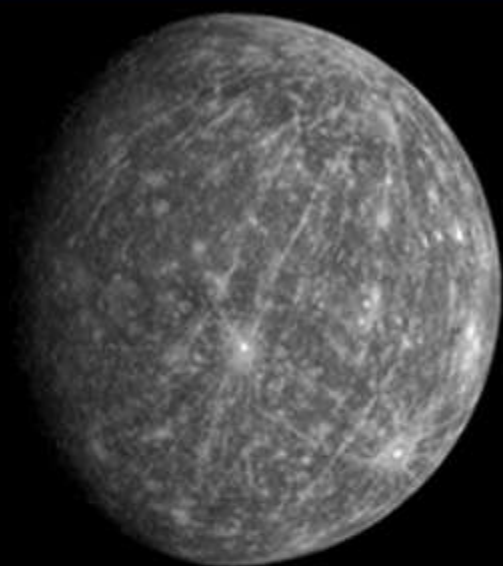
## TERRESTRIAL PLANETS

- 1 These were formed in the close vicinity of the parent star where it was too warm for gases to condense to solid particles.
- 2 The solar wind was more intense nearer to the Sun. So, it blew off lots of gas and dust from the terrestrial planets
- 3 The Terrestrial planets are smaller and their lower gravity could not hold the escaping gases.

## JOVIAN OR GAS GIANTS

- 1 Jovian planets were formed at quite a distant location.
- 2 The solar winds were not all that intense to cause similar removal of gases from the Jovian planets.





**MERCURY**



**VENUS**



**EARTH**



**MARS**



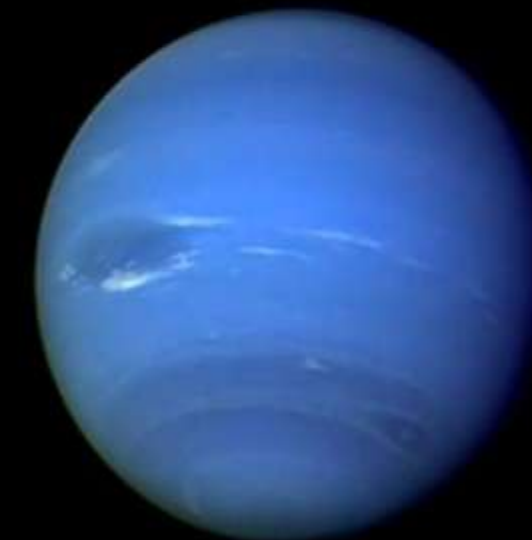
**JUPITER**



**SATURN**



**URANUS**



**NEPTUNE**





## TERRESTRIAL PLANETS

### MERCURY

- Smallest and closest to the Sun.
- Year is just 88 days.
- It is just 36 million miles away from the Sun.

### VENUS

- It is the next closest planet.
- It is considered as **Earth's twin** because of their close proximity in size, mass (weight) and density.

### EARTH

- Earth has life and we see the living things.
- Earth has a natural satellite, the Moon around 2,38,900 miles away.
- It revolves around the Earth eastward once in every 27 days.

### MARS

- It has dark patches.
- Believed by most astronomers for the possibility of some biological plant life.
- Much attention has been focussed on Mars to explore the possibility of extending man's influence on it.



# TERRESTRIAL PLANETS

Mercury



Venus



Earth



Mars







## JOVIAN PLANETS OR GAS GIANTS

### JUPITER

- The largest planet in the solar system.
- Surface is made up of many gases like Hydrogen, Helium, Methane etc.
- It's surface is very cold, probably  $-130^{\circ}\text{C}$ .
- It has got circular light and dark bands.

### SATURN

- It has three rings.
- It is second largest after Jupiter.

### URANUS

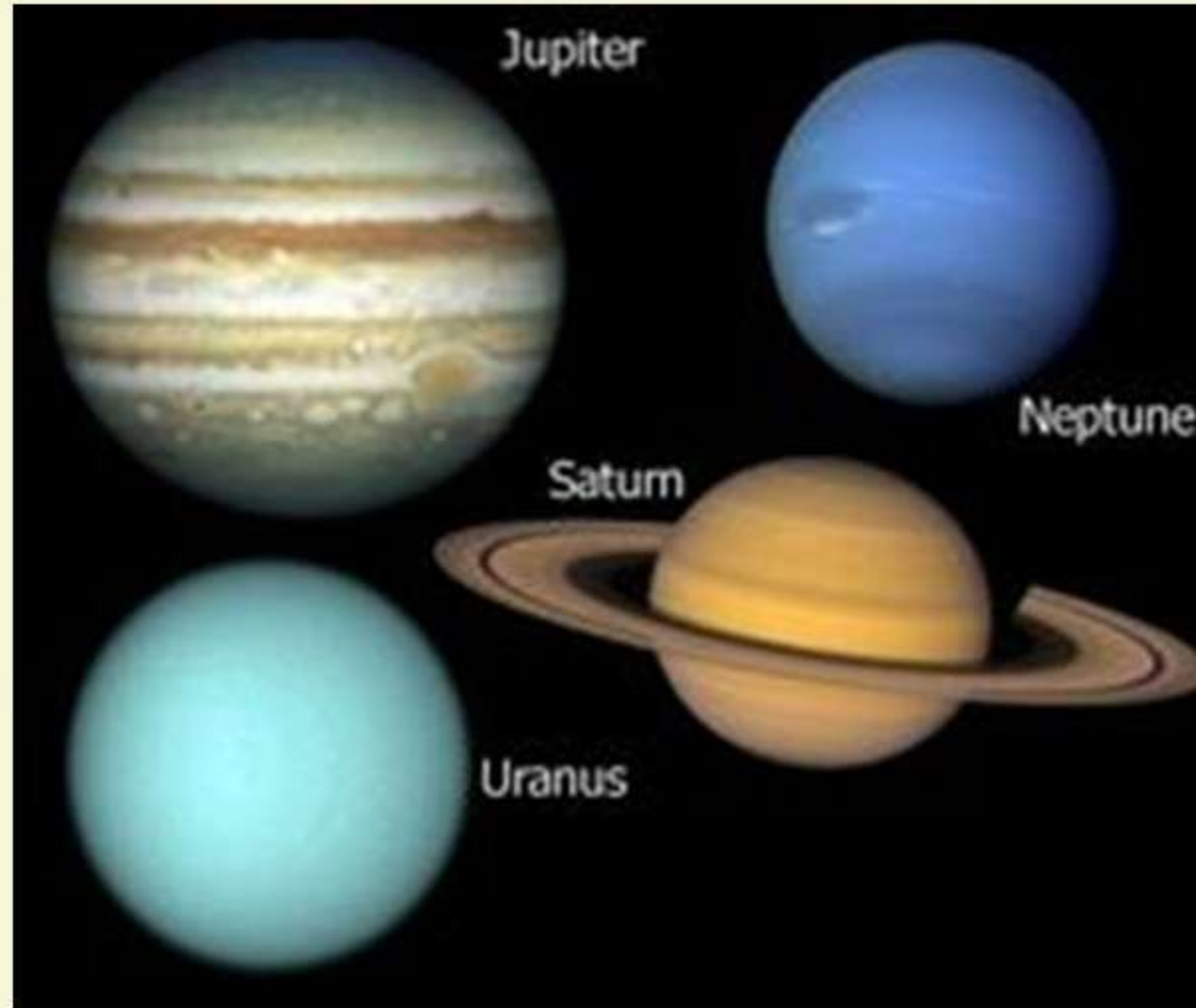
- It is known only from 18<sup>th</sup> century.
- Fifty times larger than Earth, fifteen times heavier.
- It orbits around the Sun in a clockwise direction from East to West.

### NEPTUNE

- It closely resembles Uranus.
- Probably much colder.



# JOVIAN PLANETS OR GAS GIANTS





| Mercury<br>8 <sup>th</sup>                                | Venus<br>6 <sup>th</sup>      | Earth<br>5 <sup>th</sup> | Mars<br>7 <sup>th</sup>            | Jupiter<br>1 <sup>st</sup>                | Saturn<br>2 <sup>nd</sup> | Uranus<br>3 <sup>rd</sup>              | Neptune<br>4 <sup>th</sup> |
|-----------------------------------------------------------|-------------------------------|--------------------------|------------------------------------|-------------------------------------------|---------------------------|----------------------------------------|----------------------------|
| INNER PLANETS                                             |                               |                          |                                    | OUTER PLANETS / JOVIAN PLANETS (99% mass) |                           |                                        |                            |
| Closely spaced Rocky and Terrestrial Silicates and Metals |                               |                          |                                    | Gas Giants<br>Hydrogen & Helium           |                           | Ice Giants<br>Water, Ammonia & Methane |                            |
| No Moon                                                   | No Moon                       | 1 Moon                   | 2 Moons                            | 69 Moons                                  | 61 Moons                  | 27 Moons                               | 14 Moons                   |
| Smallest                                                  | Hottest<br>@462°C             |                          |                                    | Largest & Heaviest                        |                           |                                        |                            |
|                                                           | Morning Star/<br>Evening Star | Supports Life            | Red Planet                         | Has Rings                                 | Has Rings                 | Has Rings                              | Has Rings                  |
|                                                           | Earth's Twin                  | Blue Planet              | Highest Mountain<br>(Olympus Mons) | Lord of the Heavens                       |                           |                                        |                            |
|                                                           | Brightest Planet              |                          |                                    |                                           |                           |                                        |                            |
| 88 Days                                                   | 225 Days                      | 365 Days                 | 688 Days                           | 11.9 Years                                | 29.5 Years                | 84 Years                               | 164.8 Years                |

Inner and outer planets are separated by Asteroid belt





# INTERESTING QUESTION

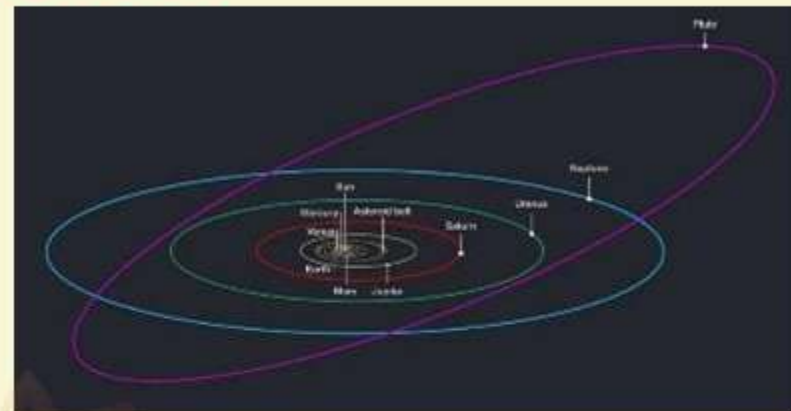


*Why Pluto was derecognized as a planet in 2006?*

## MORE INFORMATION

- ✓ Till (August 2006), Pluto was also considered a planet.
- ✓ But in a meeting of the International Astronomical Union, a decision was taken that Pluto, like other celestial objects (Ceres, 2003 UB313) discovered in recent past, were classified as “**dwarf planets.**”

*The International Astronomical Union (IAU) downgraded the status of Pluto to that of “dwarf planet”.*



## TO BECOME A PLANET

- It must be in orbit around the Sun
- It must be large enough that it takes on a nearly round shape
- It has cleared its orbit of other objects

*Pluto failed to meet the third criteria*





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